

Decision Partitions Do Not Determine Quantum Speedup

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(Dated: July 20, 2026)

A decision problem specifies which oracle instances must receive the same answer, but this partition does not by itself fix the number of queries. We use the block form of unitary discrimination to characterize exact one-query resolution. For Deutsch’s task, one query works for a controlled response unitary V exactly when -1 is an eigenvalue of V ; odd-dimensional cyclic addition therefore removes the advantage. With the standard Boolean point oracle fixed, parity needs $\lceil N/2 \rceil$ queries, while an exhaustive symmetry-reduced census shows that balanced partitions of identical size can have different costs. All 35 three-bit tasks are certified exactly. Among 6435 four-bit tasks, 395 are certified exactly; the other 6040 retain explicitly numerical status after two solver tolerances. Partitions define the required answers, whereas speedup is determined by how the oracle separates their classes.

ORACLE ACCESS AND ANSWER PARTITIONS

Two observations should be kept separate. First, query complexity is relative to an access model: a task is not intrinsically one-query quantum or classically hard. Second, even after a point-oracle model is fixed, the number and sizes of the answer classes do not determine their query cost. The first is a modeling principle; the second is the fixed-oracle claim tested below.

This separation also corrects the informal picture that a quantum query simply evaluates many classical branches in parallel [1]. A query produces one state whose relative amplitudes may be recombined. Readout depends on the distinguishability of the states associated with possible oracles, not on a branchwise list of function values. A fixed basis change leaves discrimination unchanged, whereas a different response spectrum, aggregated data, or an inaccessible environment can change the separation produced by one call [2].

Let Ω be a finite instance set and let $g : \Omega \rightarrow J$ specify the required answer. The task induces the *answer partition* (also called here the kinematic partition)

$$\Pi_g = \{C_j : j \in g(\Omega)\}, \quad C_j = g^{-1}(j). \quad (1)$$

This partition is indispensable bookkeeping, but it is not yet an access model or a complexity statement. In the query model the hidden instance labels a black-box transformation; the algorithm is not handed a state $|a\rangle$ on which the answer partition is a pre-existing observable. The operational question is whether the specified oracle and a fixed protocol can transfer the cell label to an accessible output.

More explicitly, let $\rho_a^{(T)}$ be the accessible state produced from instance a after T queries. Exact resolution means that there is one instance-independent measurement $\{P_j\}$ satisfying

$$P_j P_\ell = \delta_{j\ell} P_j, \quad \sum_j P_j = I, \quad \text{Tr}(P_{g(a)} \rho_a^{(T)}) = 1. \quad (2)$$

Consequently, supports belonging to different cells are orthogonal; no orthogonality is required within a cell. The C_j themselves remain sets of classical instances. They are not subspaces, and the answer sectors $P_j \mathcal{H}_{\text{out}}$ arise only after an oracle and protocol have encoded them. The partition therefore specifies a required zero pattern in the output distinguishability matrix rather than a ready-made physical observable.

BLOCK DISCRIMINATION WITH ONE QUERY

Consider a family of unitary oracles $\{U_a : a \in \Omega\}$. An arbitrary one-query protocol may prepare a pure state $|\Psi\rangle$ of the query system Q and a reference R , apply $U_a \otimes I_R$, and then use an instance-independent channel and measurement. Write

$$|\Psi_a\rangle = (U_a \otimes I_R) |\Psi\rangle, \quad \sigma = \text{Tr}_R |\Psi\rangle\langle\Psi|. \quad (3)$$

The output Gram matrix before the final processing is

$$G_{aa'} = \langle\Psi_a|\Psi_{a'}\rangle = \text{Tr}(\sigma U_a^\dagger U_{a'}). \quad (4)$$

Proposition 1 (block discrimination). The partition Π_g is exactly resolvable with one use of $\{U_a\}$, allowing an arbitrary reference, if and only if there is a density operator σ such that

$$\text{Tr}(\sigma U_a^\dagger U_{a'}) = 0 \quad \text{whenever} \quad g(a) \neq g(a'). \quad (5)$$

Indeed, exact discrimination requires the spans of the output states from different cells to be orthogonal, which is equivalent to the indicated pairwise inner products vanishing. Conversely, orthogonal cell spans can be measured by their support projectors. Purification shows that optimizing over σ already includes mixed probes and arbitrary ancillary systems.

Equation (5), with $\sigma \succeq 0$ and $\text{Tr} \sigma = 1$, is the direct block-discrimination form of standard perfect unitary-discrimination conditions [3, 4]; it is not a new general channel-discrimination theorem. Its role here is to

connect an answer partition to a specified oracle family. The Gram matrix need only be block diagonal according to Π_g ; states within one cell may overlap or coincide. Complete identification is the singleton-cell special case, where G must be diagonal.

For bounded error, orthogonality is replaced by a common POVM with the required success probability. Multi-query state conversion and function evaluation are characterized by adversary methods [5, 6].

Write $Q_\epsilon(\Pi; \mathcal{U})$ and $R_\epsilon(\Pi; \mathcal{U}_{\text{cl}})$ for the optimal quantum and classical worst-case numbers of calls needed with worst-case error at most ϵ , under explicitly matched access conventions, and write $Q_E \equiv Q_0$ for exact quantum complexity. Equation (5) characterizes $Q_E = 1$; it does not by itself establish an advantage. Speedup additionally requires a gap between Q_ϵ and R_ϵ for the same task and oracle interface. Classically, a deterministic query protocol partitions Ω by its possible transcripts and succeeds exactly only when that transcript partition refines Π_g . A quantum protocol need not create such an intermediate classical transcript; it changes the overlaps among the hypothesis states and measures only after the required cells have separated. The relevant resource is this rate of separation, not the number of formal components in the probe superposition.

The criterion also distinguishes a change of representation from a change of access. If

$$V_a = AU_aB \quad (6)$$

for instance-independent unitaries A, B , then $V_a^\dagger V_{a'} = B^\dagger U_a^\dagger U_{a'} B$; as the probe varies, the attainable Gram matrices are unchanged. More generally, families that exactly simulate one another with $O(1)$ oracle calls have quantum query complexities equal up to constant factors; an approximate simulation must also charge its conversion error to ϵ . Preservation of a claimed quantum-classical gap additionally requires corresponding constant-query classical simulations. Exact one-query behavior can nevertheless change when the response representation is not of the form Eq. (6).

For example, Hadamard conjugation of the answer qubit turns Eq. (7) below into the full controlled-phase action $|x, z\rangle \mapsto (-1)^{z b_x} |x, z\rangle$, so the two forms are exactly equivalent. The restricted sign oracle $P_b |x\rangle = (-1)^{b_x} |x\rangle$ is only a subroutine: $P_{b \oplus 1} = -P_b$, and the complementary truth tables differ by a global phase. Supplying a coherent control for P_b restores the missing relative phase, but coherent control is additional black-box access rather than an automatic operation on an unknown unitary [7].

DEUTSCH'S PROBLEM AND THE RESPONSE SPECTRUM

Deutsch's problem illustrates both the block criterion and this dependence. Let $b = (b_0, b_1)$ be the truth table of a Boolean function and use

$$O_b |x, y\rangle = |x, y \oplus b_x\rangle. \quad (7)$$

With the answer qubit in $|-\rangle$, phase kickback gives, up to the common answer factor, the output states on the address register:

$$\begin{aligned} |\psi_{00}\rangle &= |+\rangle, & |\psi_{11}\rangle &= -|+\rangle, \\ |\psi_{01}\rangle &= |-\rangle, & |\psi_{10}\rangle &= -|-\rangle. \end{aligned} \quad (8)$$

Thus the constant and balanced cells occupy orthogonal rays, while the two instances inside either cell are not distinguished. This is precisely a block-diagonal, but not diagonal, Gram matrix and gives the familiar one versus two exact-query separation [8].

The phenomenon is controlled by the response spectrum. Let V be any unitary and consider the controlled-response family indexed by the same four Boolean truth tables,

$$U_b^{(V)} = \sum_{x=0}^1 |x\rangle\langle x| \otimes V^{b_x}. \quad (9)$$

Theorem 1 (response spectrum). The Deutsch constant-balanced partition is exactly one-query resolvable through $\{U_b^{(V)}\}$, with arbitrary probes and references, if and only if -1 is an eigenvalue of V .

For necessity, write the probe as $|0\rangle|\phi_0\rangle + |1\rangle|\phi_1\rangle$, with $|\phi_x\rangle$ on the response and reference systems. Let $\rho_x = \text{Tr}_R |\phi_x\rangle\langle\phi_x| \succeq 0$ be its unnormalized response operator and put

$$p_x = \text{Tr } \rho_x, \quad z_x = \text{Tr}(\rho_x V), \quad p_0 + p_1 = 1. \quad (10)$$

The four cross-cell orthogonality conditions include their complex conjugates and reduce to

$$z_0 = z_0^* = -p_1, \quad z_1 = z_1^* = -p_0. \quad (11)$$

Since $|z_x| \leq p_x$, these equations force $p_0 = p_1 = 1/2$ and $z_0 = z_1 = -1/2$. Equality in the unitary expectation bound is possible only when the support of each nonzero ρ_x lies in the -1 eigenspace of V . Conversely, a -1 eigenvector provides ordinary phase kickback and Eq. (8).

Related qudit versions of Deutsch-Jozsa use higher-dimensional address and response systems [9]. The theorem above instead holds the four Deutsch instances fixed and varies the response unitary. It is also distinct from prior-distribution “useless-query” bounds, which infer quantum lower bounds from classical queries that reveal no information about a property [10].

Corollary 1 (cyclic response). For the elementary cyclic response $X_d|y\rangle = |y+1 \bmod d\rangle$, $d \geq 2$, computational-basis queries return the same Boolean point value for every d (by mapping $y \mapsto y+b_x \pmod d$ with $b_x \in \{0,1\}$), but -1 occurs in the spectrum of X_d exactly when d is even. Therefore

$$Q_E^{(d)} = \begin{cases} 1, & d \text{ even}, \\ 2, & d \text{ odd}, \end{cases} \quad D^{(d)} = 2. \quad (12)$$

In particular, reversible qutrit addition removes Deutsch's exact quantum advantage without changing its classical point values. This is a constant-factor effect, not an asymptotic separation, but it demonstrates that reversibility and blank-register behavior do not determine coherent query power.

GENERAL POINTWISE PARITY

The obstruction persists more strongly for general parity under the standard point oracle. Let $b \in \{0,1\}^N$ and $O_b|i,y\rangle = |i,y \oplus b_i\rangle$. Every pure probe, including a reference system, can be written in the answer qubit's X basis as

$$|\Psi\rangle = \sum_{i=1}^N |i\rangle (|+\rangle |\alpha_i\rangle + |-\rangle |\beta_i\rangle), \quad p_i = \|\beta_i\|^2. \quad (13)$$

The overlap between the outputs for 0^N and the odd string e_i is

$$\langle \Psi_{0^N} | \Psi_{e_i} \rangle = 1 - 2p_i. \quad (14)$$

Exact even-odd separation would require $p_i = 1/2$ for every i , while normalization gives $\sum_i p_i \leq 1$. Hence one query is possible only for $N \leq 2$, with Deutsch as the limiting case. More generally, the acceptance probability of a T -query algorithm is a polynomial of degree at most $2T$ in the input bits, whereas parity has exact degree N . Thus $T \geq N/2$, and applying the Deutsch construction to disjoint pairs gives the matching algorithm, so

$$Q_E(\text{Par}_N) = \lceil N/2 \rceil, \quad D(\text{Par}_N) = N \quad (15)$$

for point-value access [11, 12]. A binary partition can therefore remain query intensive. If instead one declares $\tilde{O}_b|y\rangle = |y \oplus \text{Par}(b)\rangle$ to be a single oracle call, parity becomes one-query resolvable classically as well as quantum mechanically. That does not defeat Eq. (15); it replaces point access by an oracle that already aggregates the desired answer.

BALANCED-PARTITION CENSUS

The preceding examples suggest that the partition is a task specification, while cost is controlled by its relation to the available queries. This can be tested exhaustively for small point oracles. Let the hidden instance be

TABLE I. Balanced three-bit tasks up to coordinate permutations, input complements, and answer exchange. The orbit sizes sum to 35.

Representative	Orbit	D	Q_E
x_1	3	1	1
$x_1 \oplus x_2$	3	2	1
Par_3	1	3	2
Maj_3	4	3	2
$x_1 \oplus x_2 x_3$	12	3	2
$\text{Sel}(x_1, x_2, x_3)$	12	2	2

$b \in \{0,1\}^N$, accessed through Eq. (7) with the address range enlarged to N , and consider every binary answer map having 2^{N-1} instances in each cell. Exchanging the answer labels gives the same unlabeled partition, so their number is

$$M_N = \frac{1}{2} \binom{2^N}{2^{N-1}}. \quad (16)$$

Coordinate permutations, independent complements of coordinates, and answer exchange preserve the point-oracle query cost. They reduce the $M_3 = 35$ partitions to six orbits and the $M_4 = 6435$ partitions to 58. Orbit sizes are restored when reporting the counts below.

For each partition we computed the optimal deterministic decision-tree depth D and sought matching bounds on the exact quantum query complexity Q_E . One-query feasibility reduces, for the Boolean response, to rational Gram constraints and was solved over the rationals. Multiquery feasibility was formulated with the Gram-matrix semidefinite program of Barnum, Saks, and Szegedy [13]. Constructive upper bounds were retained as adaptive trees whose nodes read either one bit or the parity of two bits; the latter is one Deutsch query. Such trees are useful certificates, but they are not a general characterization of exact quantum algorithms [14].

For $N = 3$ all lower and upper certificates meet exactly. Table I shows every orbit, rather than only aggregate counts. Here $\text{Sel}(c, u, v) = u$ for $c = 0$ and v for $c = 1$.

Thus 20 of the 35 balanced tasks admit an exact advantage: three by a factor of two and seventeen by a factor of $3/2$. The certificates also display the conditional structure hidden by the bare partition. For example, three-bit majority is decided by first obtaining $b_1 \oplus b_2$; if it is zero one reads b_1 , and if it is one one reads b_3 . Likewise, $F = b_1 \oplus b_2 b_3$ is decided by first reading b_2 , followed by b_1 when $b_2 = 0$ and by the Deutsch parity $b_1 \oplus b_3$ when $b_2 = 1$. Both real multilinear polynomials have degree three, so the polynomial method excludes one query and proves these two-query constructions optimal.

For $N = 4$, exact rational lower certificates and explicit trees settle 395 partitions: 4, 6, 48, 248, and 89 have $(D, Q_E) = (1, 1), (2, 1), (2, 2), (3, 2)$, and $(4, 2)$, respectively. Of these, 343 have an exact advantage. For

the remaining 43 orbit representatives, covering 6040 partitions, exact trees give $Q_E \leq 3$, while two SCS runs at requested tolerances 10^{-7} and 10^{-8} report the two-query SDP infeasible. The resulting assignments, 1536 at $(D, Q_E) = (3, 3)$ and 4504 at $(4, 3)$, are therefore *numerical candidates*, not exact lower bounds. If accepted, they give candidate totals of 4847 speedup and 1588 no-speedup partitions. Exact dual certificates are required before these totals can be stated as theorems.

Numerical SDP studies of all Boolean functions through four bits were already reported in Ref. [14]; subsequent structural work on exact two-query algorithms also uses that four-bit numerical classification [15]. We cross-validated our orbit representatives against the published NPN identifiers of Ref. [14]. After removing dummy variables, its one- and two-query entries agree with all 15 representatives certified here. For the 43 remaining representatives, its reported optimal two-query success probabilities lie between 0.900 and 0.986, consistent with our infeasibility results but still numerical. This agreement checks the enumeration and symmetry book-keeping; it does not manufacture an exact lower bound.

The narrower contribution of the census is therefore to reorganize balanced functions as unlabeled answer partitions, quotient only by symmetries that preserve point access, attach simple exact certificates where available, and keep numerical status explicit. The observed proportions are finite-model facts, not an asymptotic speedup criterion.

BERNSTEIN–VAZIRANI AND INACCESSIBLE RECORDS

Bernstein–Vazirani supplies the opposite extreme. For $a, x \in \mathbb{F}_2^n$, let $f_a(x) = a \cdot x \pmod{2}$. One phase query prepares the character state

$$|\chi_a\rangle = 2^{-n/2} \sum_x (-1)^{a \cdot x} |x\rangle, \quad \langle \chi_a | \chi_{a'} \rangle = \delta_{aa'}. \quad (17)$$

The singleton partition is therefore resolved in one query; the final Hadamards merely map $|\chi_a\rangle$ to $|a\rangle$ [16]. This orthogonality can be degraded without changing any computational-basis value returned by the oracle.

Choose $S \subseteq \{1, \dots, n\}$, $|S| = k$, and let x_S denote the corresponding substring. Consider the unitary dilation

$$W_a^S |x, y, e\rangle = |x, y \oplus a \cdot x, e \oplus x_S\rangle. \quad (18)$$

Each call receives a fresh witness in $|0^k\rangle$, which is discarded afterward. The accessible oracle is consequently the channel

$$\Phi_a^S(\rho) = \text{Tr}_E[W_a^S(\rho \otimes |0^k\rangle\langle 0^k|)(W_a^S)^\dagger]. \quad (19)$$

Basis queries still return $a \cdot x$ exactly, but the witness records selected address information and removes coherences between sectors with different x_S , as in standard

which-way interference [17]. Define the surviving answer partition

$$\Pi_S = \{C_u : u \in \mathbb{F}_2^{n-k}\}, \quad C_u = \{a : a_{\bar{S}} = u\}. \quad (20)$$

For $k < n$ and worst-case error $\epsilon < 1/2$, the matched-interface result is

$$Q_\epsilon^\Phi(\Pi_S) = 1, \quad R_\epsilon^{\text{val}}(\Pi_S) = n - k. \quad (21)$$

Here R_ϵ^{val} is the randomized classical complexity when a caller uses the same channel Φ_a^S , but prepares and reads its query and response in the computational basis; each call then returns the point value $a \cdot x$.

For the quantum part, the usual Bernstein–Vazirani input and phase kickback, followed by tracing out the witness, leave

$$\rho_a^S = 2^{-n} \sum_{x, x' : x_S = x'_S} (-1)^{a \cdot (x \oplus x')} |x\rangle\langle x'|. \quad (22)$$

Character orthogonality after the final Hadamards gives, up to a fixed reordering of coordinates,

$$H^{\otimes n} \rho_a^S H^{\otimes n} = |a_S\rangle\langle a_S| \otimes \frac{I_{\bar{S}}}{2^k}, \quad \text{Pr}(z|a) = 2^{-k} \mathbf{1}[z_{\bar{S}} = a_{\bar{S}}]. \quad (23)$$

Thus the one-query readout exactly resolves Π_S . Zero queries cannot resolve this nontrivial partition Π_S , because the output would be independent of its cell label. The classical upper bound uses the basis queries outside S . For the lower bound, even impose the favorable promise $a_S = 0$. Under the uniform distribution on $a_{\bar{S}}$, fewer than $n - k$ returned linear equations leave an affine solution space of positive dimension, so even a decoder given the complete transcript succeeds with probability at most $1/2$. The same average bound holds after including the protocol's internal randomness, and therefore some input has success at most $1/2$. Equation (21) therefore concerns the same surviving partition and the same computational-basis value interface. It does not claim that the inaccessible witness is free: if the witness were coherently available, the fixed coupling $|x, e\rangle \mapsto |x, e \oplus x_S\rangle$ could be undone without another oracle call, restoring the ordinary Bernstein–Vazirani output. The accessible oracle here is a channel, not the globally unitary dilation. Related distinctions between standard and garbage-bearing oracle access arise throughout quantum query complexity [18].

At $k = 0$, no address coordinate is recorded and Eq. (23) is the ordinary Bernstein–Vazirani output resolving the singleton partition, giving the n -versus-one identification gap. At $k = n$, the stated interference readout is uniform and Π_S has only one cell, although ordinary basis queries can still recover a . The model separates loss of the phase-interference route from loss of the classical function values themselves.

The loss also admits a continuous form. For each $i \in S$, let the two witness records have real overlap $\langle \epsilon_0^{(i)} | \epsilon_1^{(i)} \rangle = \lambda_i \in [0, 1]$, and use the product record $|\epsilon_{x_S}\rangle = \bigotimes_{i \in S} |\epsilon_{x_i}^{(i)}\rangle$. The map $|x, y, 0\rangle \mapsto |x, y \oplus a \cdot x\rangle |\epsilon_{x_S}\rangle$ is an isometry on the initialized subspace, and hence admits a unitary extension, because the address x is retained. The same calculation gives

$$\Pr_{\lambda}(z|a) = \mathbf{1}[z_{\bar{S}} = a_{\bar{S}}] \prod_{i \in S} \frac{1 + (-1)^{z_i \oplus a_i} \lambda_i}{2}. \quad (24)$$

The stated one-query readout recovers the full string with probability $\prod_{i \in S} (1 + \lambda_i)/2$. The endpoints $\lambda_i = 1$ and 0 represent, respectively, fully retained coherence and a perfectly distinguishable witness record; intermediate overlaps interpolate the visibility and success probability of the displayed Bernstein–Vazirani readout continuously. This probability characterizes that probe and Hadamard readout, not the optimal one-query strategy for the complete query–answer channel. Determining that optimum for the singleton partition of the partially coherent channel remains open.

CONCLUSION

Equation (1) specifies which hidden instances may share an answer, but it neither supplies an input state $|a\rangle$ nor selects a physical output observable. The oracle and protocol determine how rapidly states from different cells separate. The response-spectrum theorem and the contrast between point and aggregate parity show that changing access changes this separation even when the abstract task is fixed. Conversely, the finite census holds the point oracle fixed while varying every balanced target. Its exact three-bit certificates reveal advantages built from oracle-accessible, conditionally composed refinements, while the four-bit calculation marks the boundary between proof and numerical evidence.

Shor’s and Grover’s algorithms are relevant to the partition picture, but at different levels. In period finding, the answer partition groups oracle instances by their hidden period r . For each fixed instance, the function values also partition the address domain into level sets $C_t(r) = \{x : x \equiv t \pmod{r}\}$. After modular exponentiation, measuring the value register prepares a coset state, while discarding it gives the corresponding mixture. The quantum Fourier transform converts their translation symmetry into peaks near multiples of the reciprocal period. Repeated samples and continued fractions then infer r [19]. Thus the level-set partition is an operational mechanism, whereas the partition of oracle instances by r specifies the answer. This structured pair is unlike an arbitrary balanced task in the finite census.

Grover gives the complementary example. Under the promise of either no mark or exactly one mark, let

$\Omega = \{\emptyset\} \cup \{t : t \in [M]\}$. The decision partition has the two cells $\{\emptyset\}$ and $\{t : t \in [M]\}$, while the search task refines the latter into singleton cells. The marking oracle $O_t|x\rangle = (-1)^{[x=t]}|x\rangle$ supplies only a small separation per call; amplitude amplification accumulates it in $\Theta(\sqrt{M})$ bounded-error queries rather than the classical $\Theta(M)$, and this black-box scaling is optimal [20, 21]. The speedup therefore does not come from a finer partition. It comes from the symmetry of the oracle family and the coherent rotation generated by alternating the marking and diffusion reflections.

Partitions are therefore indispensable as specifications but insufficient as complexity criteria. Quantum advantage is not caused by partition coarseness or by simultaneous access to imagined branch values. It occurs when structure-matched interference separates the required answer cells in fewer calls than any classical transcript under the same interface. Superlinear exact separations for total Boolean functions are known [22], but establishing any asymptotic speedup requires a structured family of task–oracle pairs; percentages from a finite census cannot substitute for that analysis.

The author thanks Noson S. Yanofsky for discussions that motivated this work. The author used an OpenAI Codex language-model-based assistant for structural editing, consistency checks, and implementation of the finite partition census, and accepts responsibility for the final scientific content. This research was funded in whole or in part by the Austrian Science Fund (FWF), grant PIN5424624. The author acknowledges TU Wien Bibliothek for support through its Open Access Funding Programme.

Data availability.—The source code, exact certificates, numerical witnesses, generated tables, and the NPN cross-validation underlying the finite census are available in the [public supplementary archive](#) and from the author upon reasonable request.

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